

MD. FARHAN RAFIQ

Naogaon, Bangladesh | +8801717241308 | farhan.ocg@gmail.com | farhan-rafiq.github.io

EDUCATION

M.S. (Thesis) in Oceanography

Dec 2025

CGPA: 3.83/4.00 | *Shahjalal University of Science and Technology (SUST), Bangladesh*

B.Sc. in Oceanography

Dec 2023

CGPA: 3.66/4.00 | *Shahjalal University of Science and Technology (SUST), Bangladesh*

RESEARCH INTERESTS

Geographic Information System (GIS), remote sensing and earth observation, spatial data science and predictive modelling, geospatial artificial intelligence (GeoAI), hydrological and geospatial modelling, natural hazards and disaster resilience, coastal and environmental monitoring.

RESEARCH EXPERIENCE

Master's Thesis

Coastal Urban Flood Risk Modelling in Chattogram: An Integrated Machine Learning and TOPSIS Approach

- Developed an integrated urban flood-risk assessment framework, combining machine learning, multi-criteria decision analysis, and GIS-based spatial modelling.
- Trained and benchmarked RF, GBM, and SVM classifiers on a 70/30 train-test split; RF was the top-performing hazard model with a test AUC of 0.917.
- Modelled flood vulnerability using TOPSIS with AHP-derived criterion weighting.
- Produced composite flood hazard, vulnerability, and risk maps in ArcGIS, showing approximately 21% of Chattogram District falling within high to very high flood risk.

Undergraduate Research Project

Spatial and Temporal Variability of PAR, K_d_{490} , and SST in the Bangladesh EEZ (2016–2022)

- Analyzed 7 years of PAR, K_d_{490} , and SST data across the Bangladesh EEZ using SeaDAS and ArcGIS.
- Quantified coastal-offshore gradients in PAR (32.5–55 Einstein/m²/day) and turbidity (K_d_{490} : 0.01–0.33 m⁻¹), linked to land runoff.
- Tracked seasonal SST variability (22–31°C) and a gradual multi-year warming trend.

Mini Project (Remote Sensing Coursework)

Annual Variability of Chlorophyll-a and Sea Surface Temperature in the Bay of Bengal

- Processed 12 months (2020) of Ocean Color Web Chl-a and SST data using SNAP and ArcGIS.
- Found annual Chl-a ranging 0.19–0.65 mg/m³, with a seasonal shift in abundance from eastern to central/western waters.
- Linked Chl-a–SST negative-correlation periods to upwelling-driven nutrient enrichment and river discharge.

PROFESSIONAL EXPERIENCE

Research Assistant

Dec 2025 – Present

Dept. of Oceanography, SUST – Meghna River Estuary Ecohydrological Project

- Monitor ecohydrological conditions in the Meghna River Estuary, tracking hydrological and water-quality variables.
- Combine field observations with GIS-based spatial analysis to assess estuarine environmental change.

Research Intern

Mar 2025

Bangladesh Fisheries Research Institute (BFRI), Cox's Bazar, Bangladesh

- Collected phytoplankton samples from Bay of Bengal waters using a 20 µm plankton net and standard preservation techniques.
- Isolated and cultured 29 phytoplankton strains via serial dilution, preparing Guillard's F/2 media under sterile conditions.
- Ran comparative growth trials of *Skeletonema* sp. in artificial vs. natural seawater, tracking cell density over 7 days.
- Identified species morphologically using light and fluorescence microscopy (10x–40x).

Teaching Experience

Oct 2021 – Present

Academic Instructor, Sylhet, Bangladesh

- Teach Mathematics, Physics, Chemistry, Biology and Science to over 35 O-level and A-level students.
- Evaluate student progress regularly and address learning gaps.

PUBLICATIONS

Rafiq M. F., Siddique A. B. "Spatial and Temporal Dynamics of PAR, Kd₄₉₀, and SST in the Bangladesh EEZ (2016–2022)." In *Proceedings of the 18th International Conference on Engineering and Natural Sciences (ICENS 2025)*, New York, USA. ISBN: 979-8-89695-2794

CONFERENCE PRESENTATIONS

- **International Symposium on Coast, Climate and Community, Sylhet** 17 – 18 Jun 2026
Poster Presentation: "Coastal Urban Flood Risk Modelling in Chattogram: An Integrated Machine Learning and TOPSIS Approach"
- **International Conference on Engineering and Natural Sciences, Kyrenia, Cyprus** 27 – 29 Nov 2025
Oral Presentation (online): "Spatial and Temporal Dynamics of PAR, Kd₄₉₀, and SST in the Bangladesh EEZ (2016–2022)"

TECHNICAL SKILLS

- **GIS & Remote Sensing:** ArcGIS Pro, ArcMap, QGIS, SNAP, SeaDAS, ODV, Delft3D, Google Earth Engine
- **Programming Languages:** Python (Pandas, NumPy, Scikit-learn, Matplotlib), R
- **Oceanographic Instruments:** CTD, ADCP, Water Multi-parameter, AlgaeTorch
- **Data Analysis & Visualization:** Microsoft Excel, Google Sheets, Power BI
- **Research & Writing Tools:** LaTeX, Mendeley, Microsoft Word, Google Docs
- **Survey Tools:** Kobo Toolbox, Microsoft Forms, Google Forms
- **Graphic Design:** Canva Pro, Adobe Illustrator, Microsoft PowerPoint

HONORS & AWARDS

National Science and Technology Fellowship 2024 – 2025
Granted by The Ministry of Science and Technology, Government of Bangladesh

CERTIFICATIONS & TRAINING

- **Workshop on Affordable Coastal Observation** – Dept. of Oceanography, SUST
- **Going Places with Spatial Analysis** – Esri
- **Spatial Analysis and Satellite Imagery in a GIS** – University of Toronto, Coursera
- **Databases and SQL for Data Science with Python** – IBM, Coursera
- **Data Analysis with Python** – IBM, Coursera
- **Preparing Data for Analysis with Microsoft Excel** – Microsoft, Coursera
- **Cartography** – Esri

LANGUAGE PROFICIENCY

- **English:** Proficient
- **Bangla:** Native

CO-CURRICULAR ACTIVITIES

Volunteer, International Symposium on Marine Resource Management, SUST 24 – 25 Mar 2023
• Supported symposium logistics, session coordination, and participant facilitation.

REFERENCES

1. **Dr. Subrata Sarker** | Associate Professor and Head, Oceanography, SUST | subratasrk-ocg@sust.edu
2. **Abu Bokkar Siddique** | Assistant Professor, Oceanography, SUST | absiddique-ocg@sust.edu
3. **Muhammad Mizanur Rahman** | Assistant Professor, Oceanography, SUST | mmrahman-ocg@sust.edu